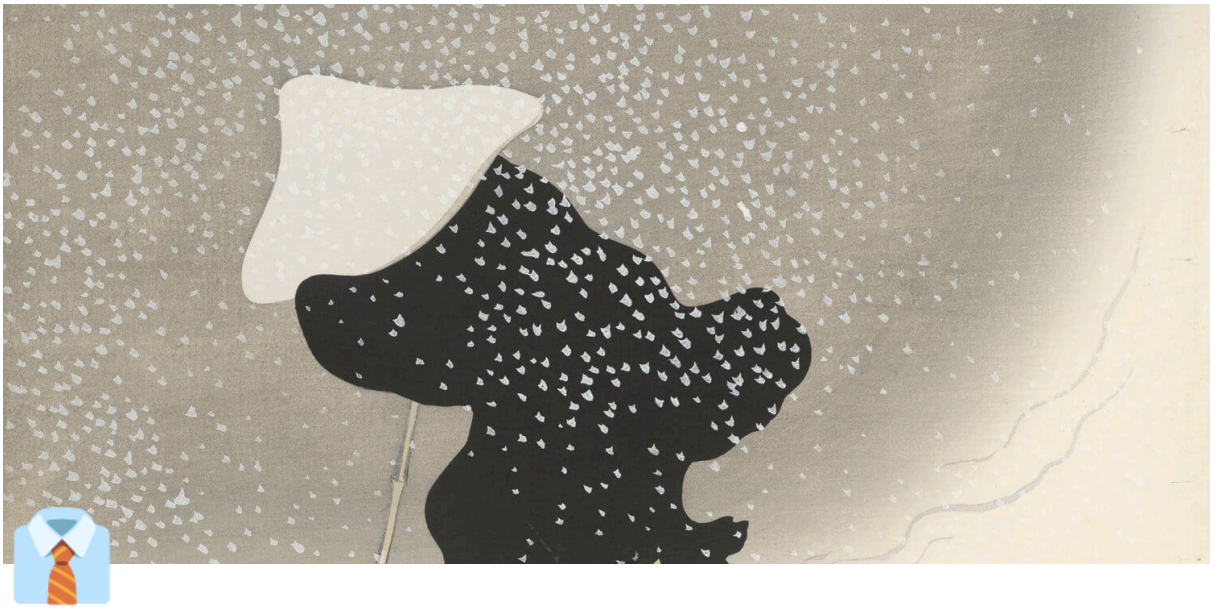


Abstract Algebra/Modern Algebra

Groups

An example of a group: Addition modulo 3

Is g = <{0, 1}, XOR> a group?

Abelian groups

Theorem: A group has only 1 identity element

Subgroups, Semigroup and Monoids

Theorem: If H and J are subgroups of G, then H INTERSECTION J is a subgroup of G

Semigroup

Monoid

Example: Monoid or Semigroup?

Homomorphism

Groups

A group (g) consists of a set G and operation o.

g = <G, o>

A group satisfies four axioms

1. G is **closed under** o

$$a, b \in G, a \ o \ b \in G$$

2. o is **associative**

3. G contains an **identity** element e

$$e \ o \ a = a \ o \ e = a$$

4. for every element in the group, there better be an **inverse**

$$\forall g \in G, \text{g has an inverse } g^{-1}$$

$$g \ o \ g^{-1} = g^{-1} o \ g = e$$

An example of a group: Addition modulo 3

$$G =< \mathbb{Z}_3, + >$$

+	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

1. closed, <Z_3> = {0, 1, 2}
2. associative, yes
3. identity, e+, 0 is the identity element
4. inverse, every element has an inverse:
    a. 0 inverse = 0
    b. 1 inverse = 2
    c. 2 inverse = 1

the group is also communicative (order don't matter)

⇒ Abelian Group

Is g = <{0, 1}, XOR> a group?

XOR	0	1
0	0	1
1	1	0

1. closed {0, 1}
2. associative? i think so
    a. to check: (1 XOR 0) XOR 0 = 1 XOR (0 XOR 0)
        i. 1 = 1, yes

3. identity element? = 0
4. inverse
    a. 0^-1 = 0
    b. 1^-1 = 1

Satisfies all 4 axiom, therefore a group.

Abelian groups

Groups that are commutative. = > Abelian groups

Example:

- <R, *>
    - 3 * 2 = 2 * 3
- <U, INTERSECTION>
    - A N B = B N A
- <Z, +>
    - 2 + 3 = 3 + 2

Theorem: A group has only 1 identity element

Proof:

$$let\ G =< G, o > \text{ be a group}$$
$$Assume\ \text{e, e' are idenetities and e is NOT e' i.e.}$$
$$e \neq e$$
$$e\ o\ a = a\ o\ e = a$$
$$e'\ o\ a = a\ o\ e' = a$$
$$e = e'\ o\ e$$
$$= e'\ o\ e$$
$$= e'$$
$$e = e'$$
$$\text{this is a contradiction}$$

Subgroups, Semigroup and Monoids

g' is a subgroup of g if and only if

1. g' is a group

$$2. \forall a, b \in g', \text{a, b} \in g$$

example:

$$g =< \mathbb{Z}, + > \quad g' =< 2\mathbb{Z}, + >$$
$$\{-2, -1, 0, 1, 2..\} \quad \{-2, 0, 2\}$$
$$\text{g, g' are groups, } 2\mathbb{Z} \subseteq \mathbb{Z}$$
$$g' \text{is a subgroup of } g$$

Theorem: If H and J are subgroups of G, then H INTERSECTION J is a subgroup of G

Proof:

$$(i) a, b \in H \cap J...$$
$$a, b \in H \; and \; a, b \in J$$
$$a \; o \; b \in H \; and \; a \; o \; b \in J$$
$$=> a \; o \; b \in H \cap J$$
$$(ii) e \in H, e \in J... => e \in H \cap J$$
$$(iii) a \in H \cap J... =>$$
$$(a \in H => a^{-1} \in H, a \in J => a^{-1} \in J) =>$$
$$a^{-1} \in H \cap J$$

associativity is left out ^

Semigroup

H = <H, o> is a semigroup if and only if:

1. H is closed under o
2. o is associative

Monoid

A semigroup with an identity;

H = <H, o> is a monoid if and only if:

1. H is closed under o
2. o is associative
3. H has an identity element.

Monoid is a set equipped with an associative binary operation and an identity element. e.g. the nonnegative integers with addition form a monoid, the identity element being 0.

Or you can think of Monoids as groups without inverses.

Example: Monoid or Semigroup?

$$< 2\mathbb{Z}, * >$$

only have to check for “does an identity element”

$$2\mathbb{Z} = \{... -2, 0, 2, 4...\}$$
$$\text{for just } \mathbb{Z}\ e_x = 1, \text{but for } 2\mathbb{Z}, \text{there is no identity}$$
$$\text{this is a semigroup}$$

$$< 2\mathbb{Z} + 1, * >$$

for does identity element exist

$$2\mathbb{Z} + 1 = \{... -1, 1, 3, 5...\}$$
$$\text{odd * odd} = \text{odd} => closed$$
$$\text{multiplication are associatve}$$
$$e_x = 1, \text{it is still in the set of odd integers}$$
$$\text{this is a monoid}$$

Homomorphism

Lets imagine there exists 2 groups: Group G1, Group G2.

How could we compare the two? i.e. how could we determine how different or similar they are.

Consider the group of integer Z, and the group of integers mod 2, the operation is both group are +.

The first group of integers is a infinite group.

The second group is a finite group with only 2 elements.

The first group of integer can be broken up into 2 sets, the set of even integers and the set of odd integers.

> if you add a even integer with another even integer you get even integer. i.e. even + even = even
>
> same logic can be applied to:
>
> even + odd = odd and
>
> odd + even = odd
>
> odd + odd = even.

We can use the above logic to do the same to the second group:

> 0 + 0 congruent to 0 (mod 2)
>
> 0 + 1 congruent to 1 (mod 2)
>
> 1+ 0 congruent to 1 (mod 2)
>
> 1 + 1 congruent to 0 (mod 2)

A similarity can be noticed, as we replace 0 as even and 1 as odd, they are essentially the same.

Using more complex mathematical terms we can write the above:

$$f : \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$$

the function is pretty simple:

if the integer is even, map it to 0,

if the integer is odd, map it to 1.

Can we reverse this process?

i.e. Is it possible to create a reverse function that encodes our observation that evens and odds behaves like the group of integers mod 2?

The answer to that answer is: NO.

consider:

$$g : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}$$

we can send 0(mod 2) to some integer x

and 1 to some integer y

i.e.

$$g : 0(mod\ 2) \to x$$
$$g : 1(mod\ 2) \to y$$

The integer mod 2:

since

$$0 + 0 \equiv 0(mod\ 2)$$

we want

$$x + x = x$$

in the Integers

This is because as we are comparing groups, the behaviour of elements in the output should be similar to the behaviour of the input.

Simplify the x + x = x, we get x = 0

so

$$g : 0(mod\ 2) \to 0$$
$$g : 1(mod\ 2) \to y$$

next

$$1 + 1 \equiv 0(mod\ 2)$$

we want:

$$y + y = 0$$

from this result, we are able to derive y = 0.

so

$$g : 0(mod\ 2) \to 0$$
$$g : 1(mod\ 2) \to 0$$

The function g is trivial, it sends everything in integer mod 2 to 0.

In this direction, we lose the group similarity between the evens and the odds and the integers mod 2.

This example above demonstrates that direction of mapping matters.

Let's consider another example

$$Group\ 1 : \mathbb{Z}/4\mathbb{Z} \text{ -> integer mod 4 and the operation: } +$$
$$Group\ 2 : \{1, -1, i, -i\} \text{-> the set of foreign number and the operation: } *$$

To compare them, we will take a look at the Cayley table(wtf) for both groups.

So, the first group is said to be "The group of integer mod 4 under addition."

The Cayley table for the first group:

+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

The Cayley table for the second group:

*	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i

i	i	-i	-1	1
-i	-i	i	1	-1

From the Cayley table,

we notice that:

- they are the same size

But do they behave similarly? Or are they different and unrelated to each other?

We can answer the question visually by using the table above.

+	0	1
0	**0**	1
1	1	2
2	2	3
3	3	0

The identity element ^ $\Rightarrow$ 0

*	1	-1
1	**1**	-1
-1	-1	1
i	i	-i
-i	-i	i

The identity element ^ $\Rightarrow$ 1, the help us to see the pattern, we can swap -1 and i.

+	0	1
0	**0**	1
1	1	2
2	2	3
3	3	0

*	1	i
1	**1**	i
i	i	-1
-1	-1	-i
-i	-i	1

The coloured elements have a similar pattern. Hell they the fucking same.

They're identical groups. They just use a different elements and a different operation, but other than that, the groups are equivalent.

These two group are **isomorphic**. (equal form)

Generalisation

Let's consider two groups:

$$Group\ 1 : (G, *)$$
$$Group\ 2 : (H, \diamond)$$

These two groups can be whatever you want, communitive, non-communitive...

Pick 2 elements in G: x, y $\in G$

if we combine x and y, we get a third element in G i.e.

$$x * y \in G$$

we can call this x "times" for now but could be something completely different

for group 1 and group 2 to have similar group behaviour, x, y and x * y in G must correspond to the element in both groups i.e.

Group 1: G

*	…	Y
…		
X		X * Y
…		

Group 2: H

operation	…	f(Y)
…		
f(X)		f(X * Y)
…		

the mathematical way to write this way is:

$$f : G \to H$$

other key observation (that I did not make), is in the second table

$$f(x) \diamond f(y)$$

should also be in it.

i.e.

operation	…	f(Y)	…
…			
f(X)		f(X * Y) f(x) operation f(y)	
…			

i.e. f(x) operation f(y) = f(x * y)

We now have derive a way to compare two groups by using a function:

$$f : G \to H$$
$$f(x * y) = f(x) \diamond f(y)$$

we call F a group homomorphism, or just **homomorphism** for short.

Note: the operation on the left are operations from G, while the operations on the right are operations form H.

Example

Let's consider

$$f : \mathbb{Z} \to \mathbb{Z}$$
$$f(x) = 2x$$
$$\text{Group operation: } +$$

To check if it's homomorphism, we only have show that:

$$f(x+y) = f(x) + f(y)$$

from the definition given above:

$$2(x+y) = 2(x) + 2(y)$$
$$\text{this will be true through distribution}$$

therefore f is a homomorphism,

Note:

the output of f is a set of even integers

the even integers are a subgroup of the integers, i.e.

$$2\mathbb{Z} \leq \mathbb{Z}$$